



OPERATING SYSTEMS

UE24CS242B

Introduction, Computer System Organization

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Department of Computer Science & Engineering

OPERATING SYSTEMS

Slides Credits for all the PPTs of this course



- The slides/diagrams in this course are an **adaptation, combination, and enhancement** of material from the following resources and persons:
1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
 2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
 3. Some presentation transcripts from A. Frank – P. Weisberg
 4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

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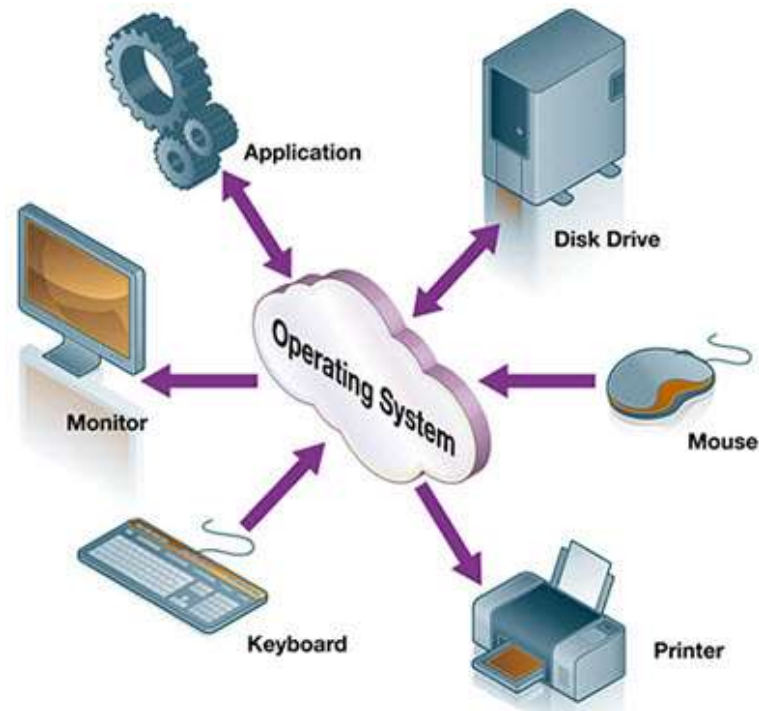
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Need for Operating System

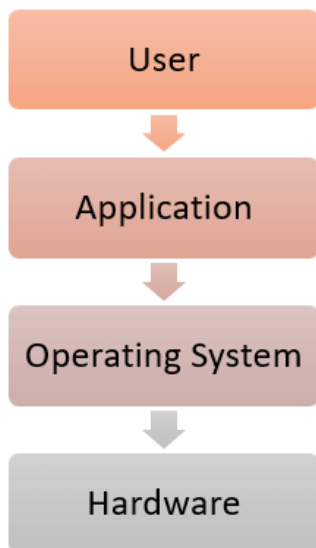


If OS is not developed then how will the application developers access the hardware?

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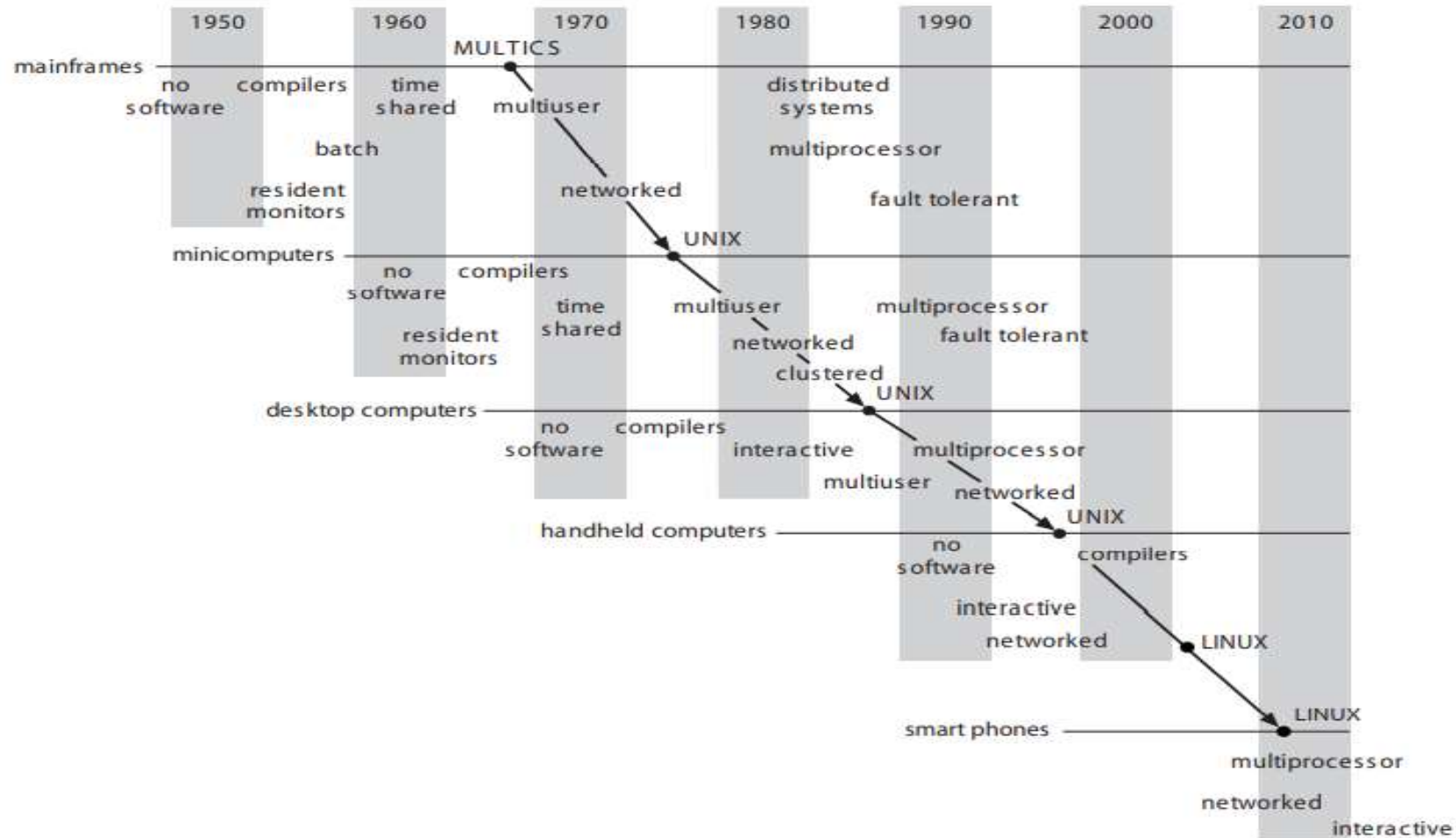
General Definition

- An Operating System is a program that acts as an intermediary between a user of a computer and the computer hardware
- It provides a user-friendly environment in which a user may easily develop and execute programs. Otherwise, hardware knowledge would be mandatory for computer programming.



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Genesis



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Operating System Goals

- Execute user programs and make solving user problems easier
- Make the computer system convenient to use
- Use the computer hardware in an efficient manner
- Manage resources such as
 - Memory
 - Processor(s)
 - I/O Devices



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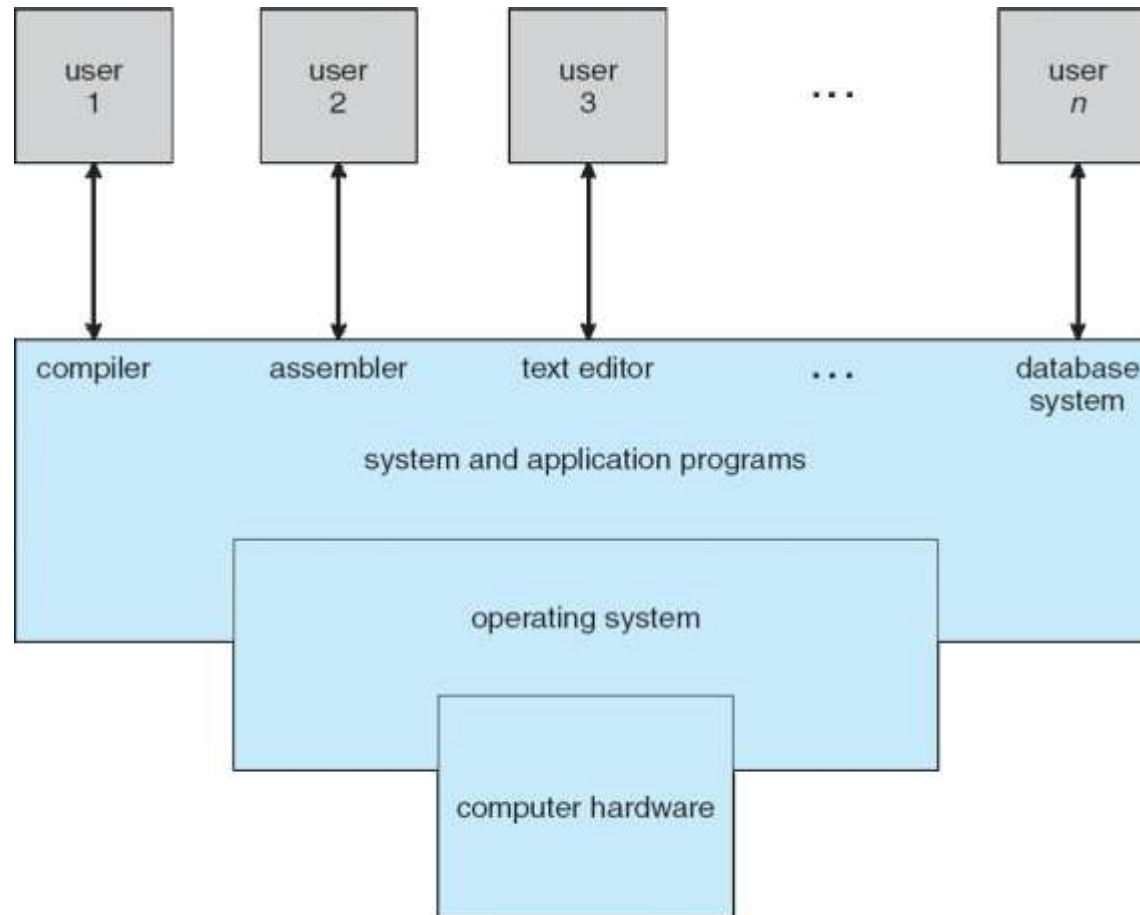
Why Study Operating Systems?

- Only a small percentage of computer practitioners will be involved in the creation or modification of an operating system.
- However almost all code runs on top of an operating system, and thus knowledge of how operating systems work is crucial to **proper, efficient, effective, and secure programming**.
- Understanding the fundamentals of operating systems



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Four Components of a Computer System (Abstract view)



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Computer System Structure



- Hardware – provides basic computing resources
 - 4 CPU, memory, I/O devices
- Operating system
 - 4 Controls and coordinates use of hardware among various applications and users
- Application programs – define the ways in which the system resources are used to solve the computing problems of the users
 - 4 Word processors, compilers, web browsers, database systems, video games
- Users
 - 4 People, machines, other computers

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What Operating Systems Do

- Depends on the point of view user and system
- Users want convenience, **ease of use** and **good performance**
 - Don't care about **resource utilization**
- But shared computer such as **mainframe** or **minicomputer** must keep all users happy.
 - Maximize resource utilization.
 - Available CPU time, memory, and I/O are used efficiently and that no individual user takes more than her fair share



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What Operating Systems Do

- Users of dedicated systems such as workstations have dedicated resources but frequently use shared resources from servers.
 - resources like file, compute, and print servers are shared.
 - operating system is designed to compromise between individual usability and resource utilization
- Handheld computers are resource poor, optimized for usability and battery life.
- Some computers have little or no user interface, such as embedded computers in devices and automobiles



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System View

- OS is a **resource allocator**
 - Manages all resources
 - Decides between conflicting requests for efficient and fair resource use
- OS is a **control program**
 - Controls execution of programs to prevent errors and improper use of the computer



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Defining Operating System

- The OS has many roles and functions
- The fundamental goal of computer systems is to execute user programs and to make solving user problems easier.
- The common functions of controlling and allocating resources are then brought together into one piece of software: the **operating system**
- “The one program running at all times on the computer” is the **kernel**.
- Everything else is either
 - a system program (ships with the operating system) , or
 - an application program.

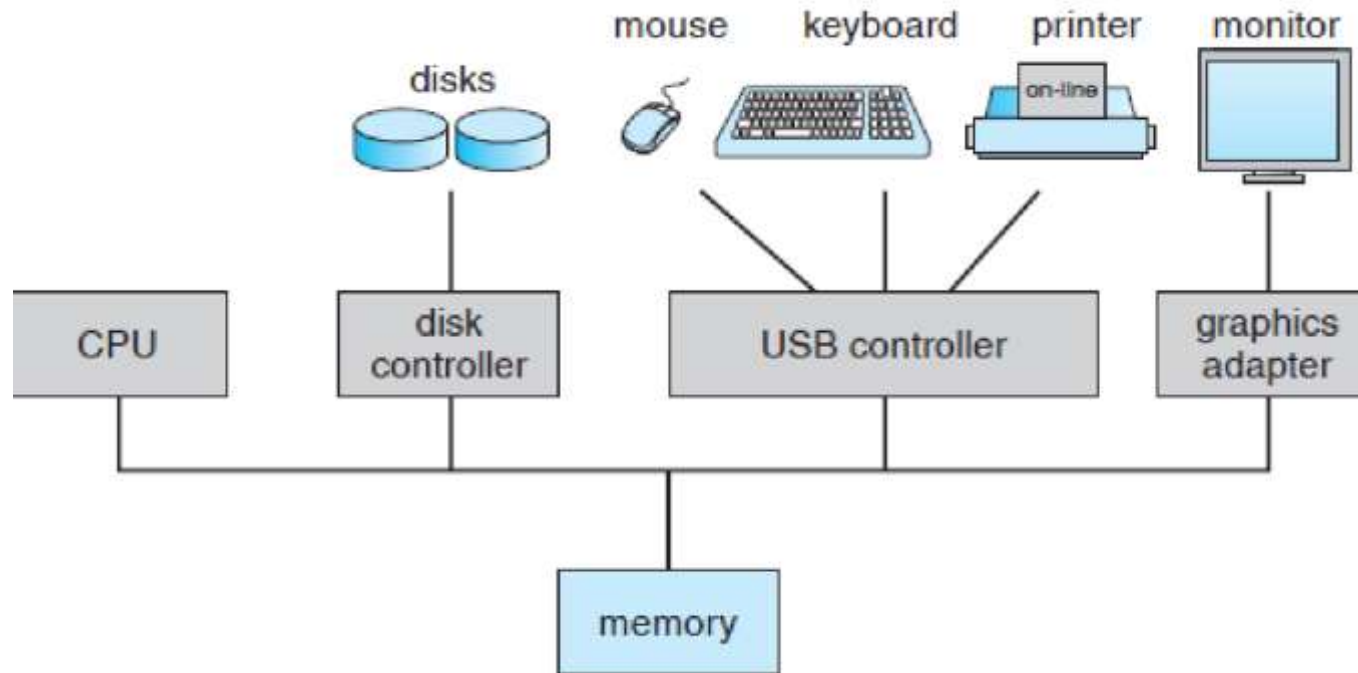




- Computer-system consists of,
 - One or more CPUs, device controllers connect through common bus providing access to shared memory
 - The CPU and the device controllers can execute concurrently, competing for memory cycles.
 - memory controller is provided to synchronize access to the memory.

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Computer System Organization



A modern computer system

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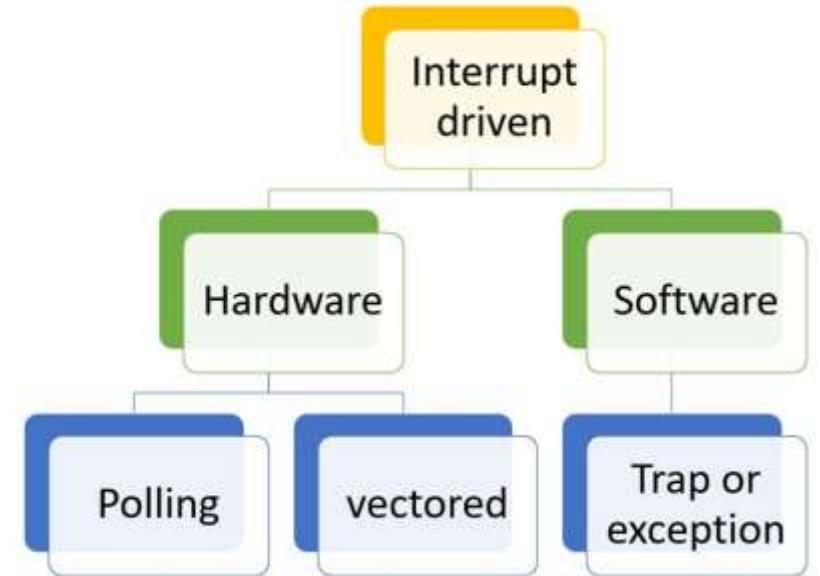
Computer System Operation

- I/O devices and the CPU can execute concurrently
- Each device controller is in charge of a particular device type
- Each device controller has a local buffer
- Each device controller has registers for action (like “read character from keyboard”) to take
- CPU moves data from/to main memory to/from local buffers
- I/O is from the device to local buffer of controller
- Device controller informs CPU that it has finished its operation by causing an **interrupt**

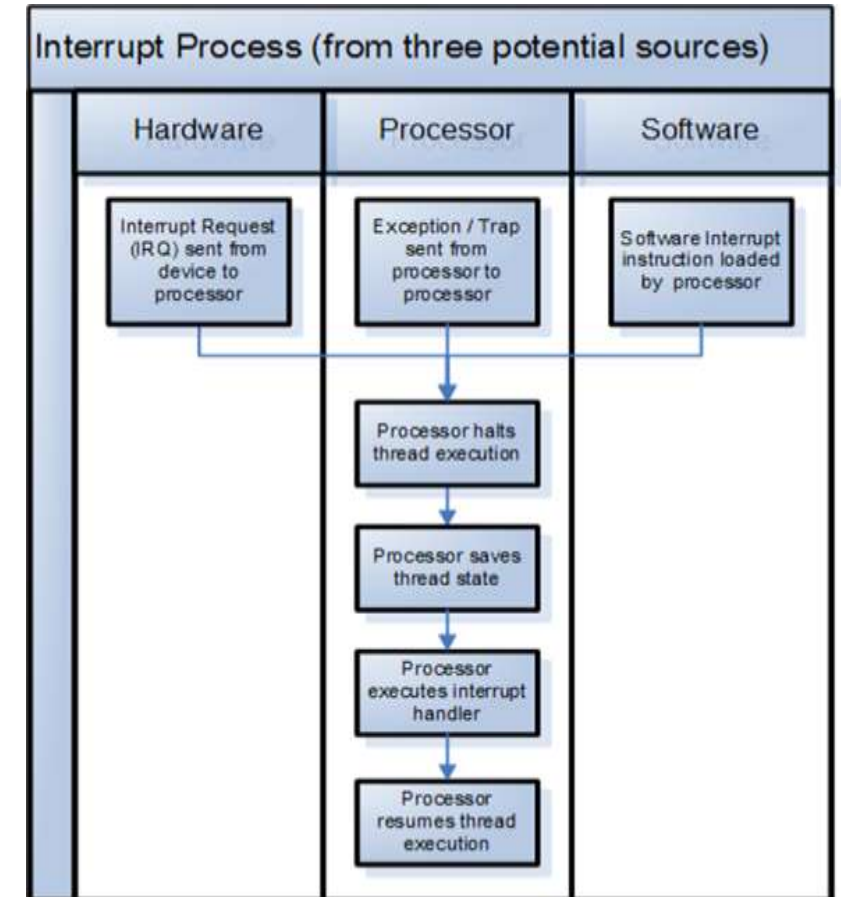


- When the system is booted, the first program that starts running is a Bootstrap.
- It is stored in read-only memory (ROM) or electrically erasable programmable read-only memory (EEPROM).
- Bootstrap is known by the general term firmware, within the computer hardware.
- It initializes all aspects of the system, from CPU registers to device controllers to memory contents.
- The bootstrap program must know how to load the operating system and how to start executing that system.
- The bootstrap program must locate and load into memory the operating system kernel.
- The first program that is created is **init**, after the OS is booted. It waits for the occurrence of event.

- An operating system is **interrupt driven**
- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector**, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A **trap** or **exception** is a software-generated interrupt caused either by an error or a user request

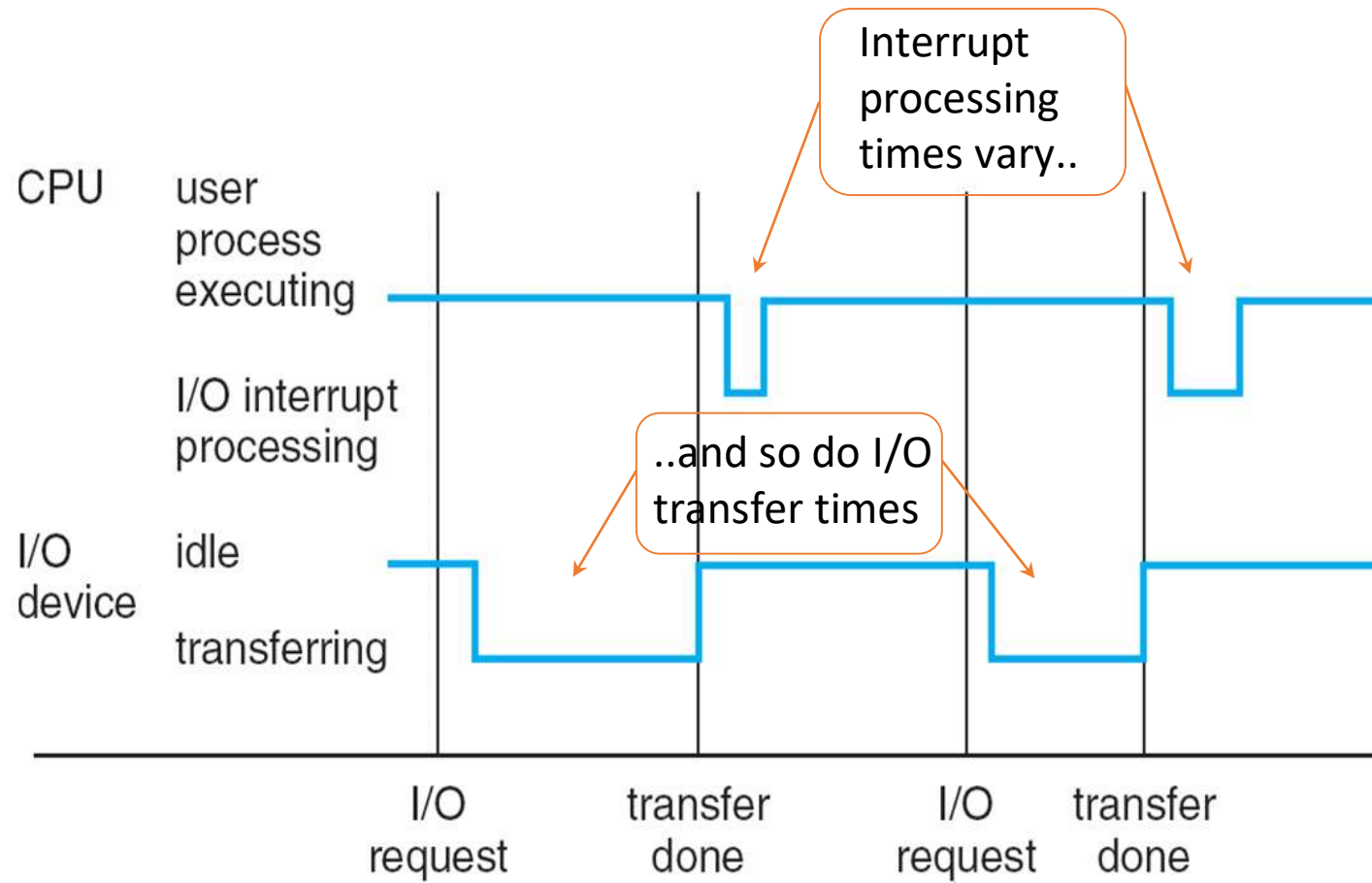


- The operating system saves the state of the CPU by storing registers and the program counter
- Determines which type of interrupt has occurred:
 - polling
 - vectored interrupt system
- Separate segments of code determine what action should be taken for each type of interrupt



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Interrupt Timeline for a single process doing output



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Storage Structure

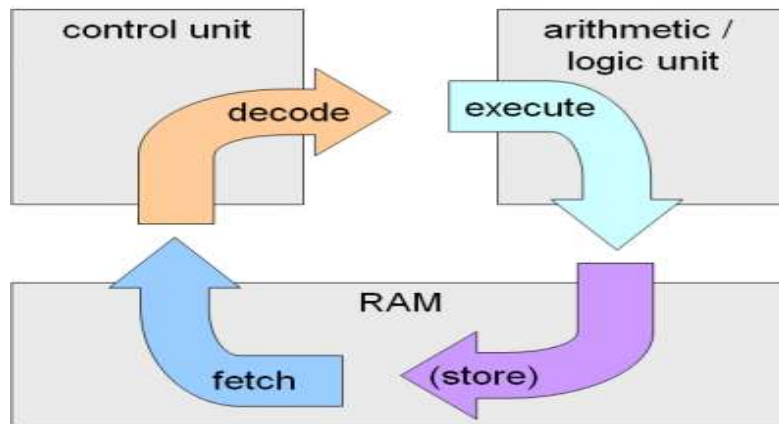
- Main memory – only large storage media that the CPU can access directly (**Random access memory** and typically **volatile**)
 - Implemented with semiconductor technology called DRAM
- Computers use other forms of memory like ROM, EEPROM
- Smart phones have EEPROM to store factory installed programs.



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Program Execution Model

- Typical instruction execution



- The processor **fetches** instructions from memory, **decodes** and **executes** them.
- The Fetch, Decode and Execute cycles are repeated until the program terminates.

Image source : https://computersciencewiki.org/index.php/The_machine_instruction_cycle

- This is called the **Von Neumann** model of computing.

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Storage Structure

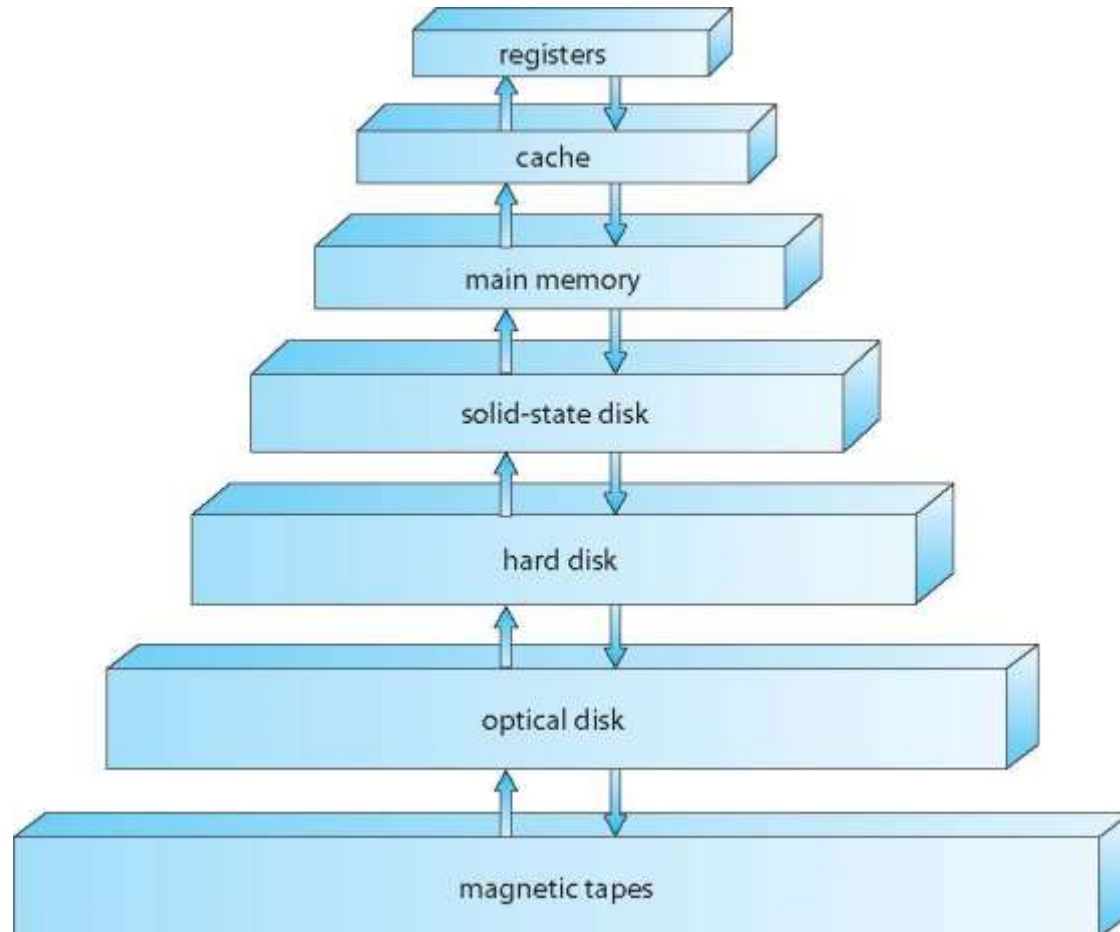
- Secondary storage – extension of main memory that provides large **nonvolatile** storage capacity
- Hard disks – rigid metal or glass platters covered with magnetic recording material
 - Disk surface is logically divided into **tracks**, which are subdivided into **sectors**
 - The **disk controller** determines the logical interaction between the device and the computer
- **Solid-state disks** – faster than hard disks, nonvolatile
 - Various technologies and becoming more popular
 - Flash memory used in camera's PDA's



- Storage systems organized in hierarchy
 - Speed
 - Cost
 - Volatility
- **Caching** – copying information into faster storage system; main memory can be viewed as a cache for secondary storage

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Storage-Device Hierarchy

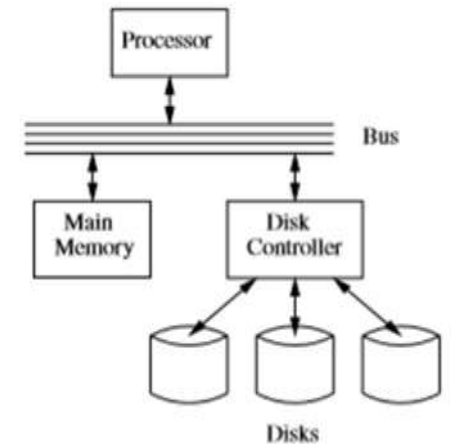


- Important principle, performed at many levels in a computer (in hardware, operating system, software)
- Information in use copied from slower to faster storage temporarily
- Faster storage (cache) checked first to determine if information is there
 - If it is, information used directly from the cache (fast)
 - If not, data copied to cache and used there
- Cache smaller than storage being cached
 - Cache management important design problem

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I/O Structure

- Storage is a type of I/O device
- A large portion of operating-system code is dedicated to managing I/O
 - As reliability and performance of a system is the main concern.
- General-purpose computer system consists of CPUs and multiple device controllers that are connected through a common bus.
- Each device controller is in charge of a specific type of device.
- **Device Driver** for each device controller to manage I/O
 - Provides uniform interface between controller and kernel
- **Small Computer-Systems Interface (SCSI)** controller enables to connect more devices.



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I/O Structure

- A device controller maintains some local buffer storage and a set of special-purpose registers.
- The device controller is responsible for moving the data between the peripheral devices.



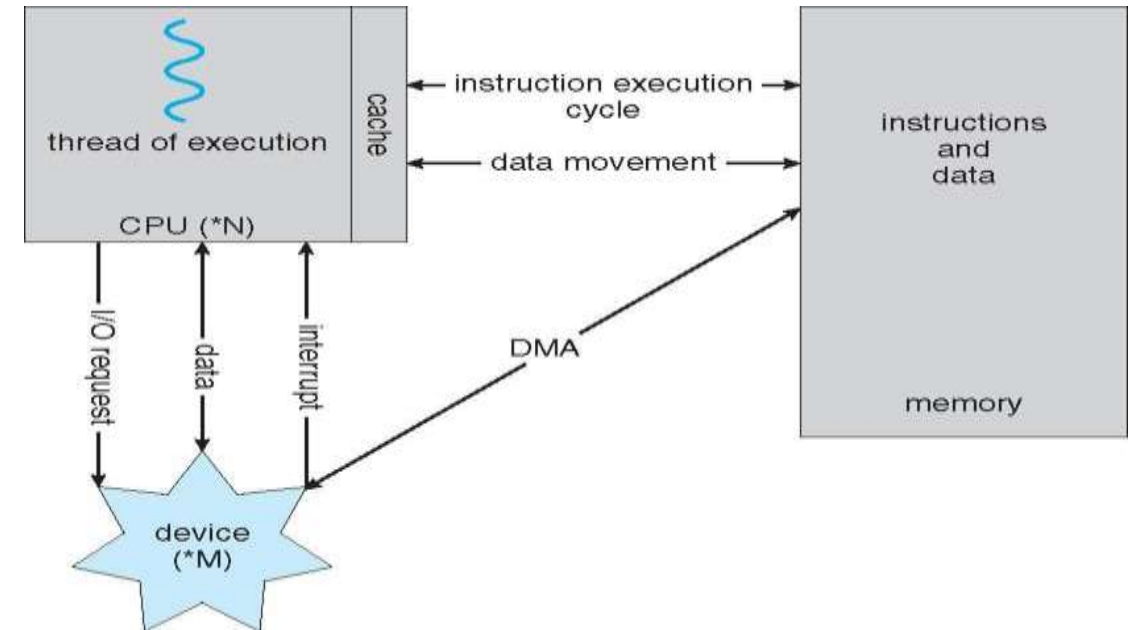
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I/O Structure



- After I/O starts, control returns to user program only upon I/O completion
 - Wait instruction idles the CPU until the next interrupt
 - Wait loop (contention for memory access)
 - At most one I/O request is outstanding at a time, no simultaneous I/O processing
- After I/O starts, control returns to user program without waiting for I/O completion
 - **System call** – request to the OS to allow user to wait for I/O completion
 - **Device-status table** contains entry for each I/O device indicating its type, address, and state
 - OS indexes into I/O device table to determine device status and to modify table entry to include interrupt.

- Used for high-speed I/O devices able to transmit information at close to memory speeds
- Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention
- Only one interrupt is generated per block, rather than the one interrupt per byte





THANK YOU

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